



PowerPoint Presentation by
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MANAGEMENT ACCOUNTING

8th EDITION

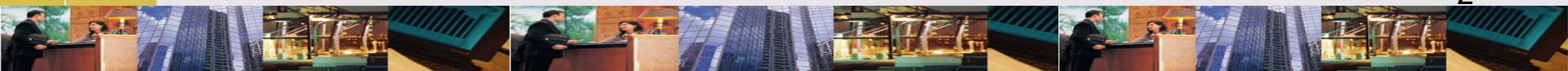
BY

HANSEN & MOWEN

8 BUDGETING FOR PLANNING & CONTROL

LEARNING OBJECTIVES

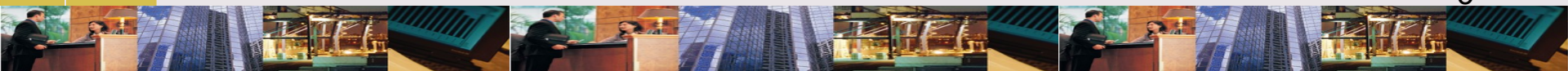
After studying this chapter, you should be able to:



LEARNING OBJECTIVES

1. Discuss budgeting & its role in planning, control, & decision making.
2. Define & prepare a master budget, identify its major components, & outline the interrelationships of its various components.

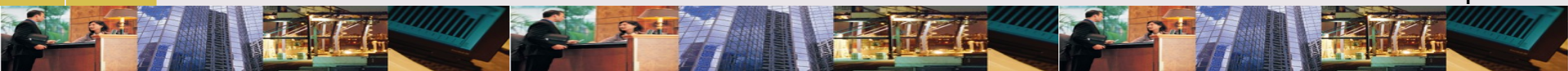
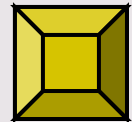
Continued



LEARNING OBJECTIVES

3. Describe flexible budgeting, & list the features that a budgetary system should have to encourage managers to engage in goal-congruent behavior.
4. Explain how activity-based budgeting works.

*Click the button to skip
Questions to Think About*



QUESTIONS TO THINK ABOUT:

Dr. Jones, DDS

Why did Dr. Jones fire his bookkeeper? Were his financial problems her fault? Why or why not?

JONES



QUESTIONS TO THINK ABOUT:

Dr. Jones, DDS

How would a formal budgeting system help Dr. Jones get out of his financial difficulties?

JONES



QUESTIONS TO THINK ABOUT:

Dr. Jones, DDS

Many small businesses do not budget, reasoning that they are small enough to mentally keep track of all revenues & expenditures. Comment on this idea.

JONES



QUESTIONS TO THINK ABOUT:

Dr. Jones, DDS

Do you budget? Explain why you do or do not?

JONES



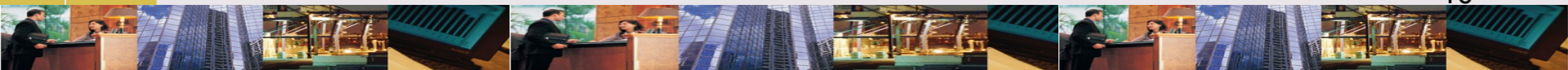
LEARNING OBJECTIVE

1

Discuss budgeting & its role in planning, control, & decision making.

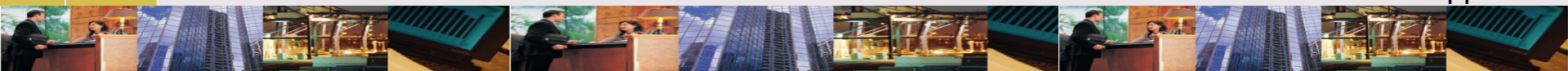
PLANNING: Definition

Looking **ahead** to see what actions should be taken to realize particular goals.



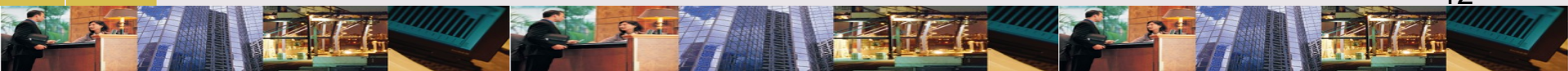
CONTROL: Definition

Looking **backward**, determining what actually happened & comparing it with previously planned outcomes.

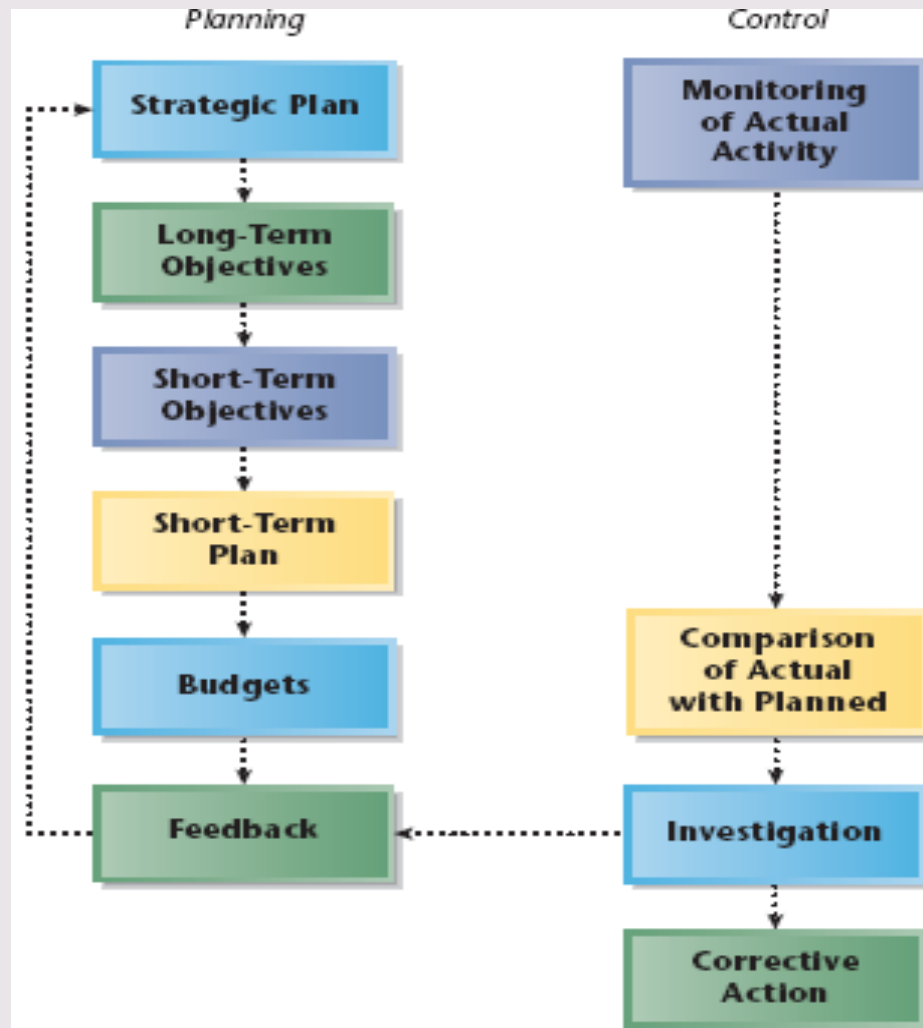


Where do budgets fit into planning & control?

Budgets are financial plans for the future, identifying objectives & the actions needed to achieve them.



PLANNING, CONTROL & BUDGETS



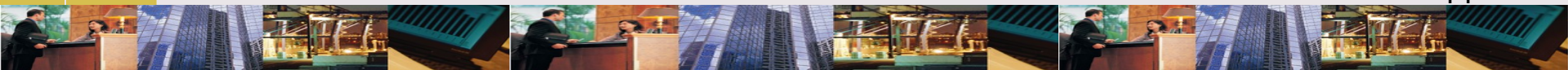
Budgets provide feedback for investigation on achieving the strategic plan.

EXHIBIT 8-1

ADVANTAGES OF BUDGETING

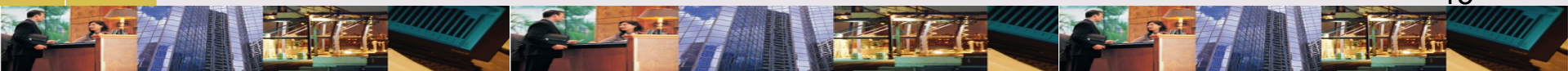
A budgetary system provides the following advantages:

- * Forces managers to plan
- * Provides information that can be used to improve decision making
- * Provides a standard for performance evaluation
- * Improves communication & coordination



BUDGETS

- * Improve decision making
- * Set standards
 - * Compare actual to budgeted results
- * Communicate & coordinate



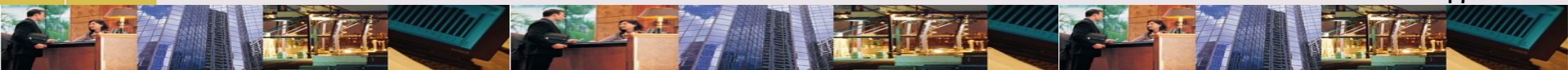
LEARNING OBJECTIVE

2

Define & prepare a master budget.

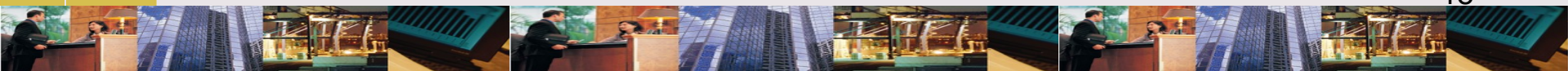
MASTER BUDGET: Definition

Comprehensive financial plan
for organization as a whole.



What is a “continuous” budget?

A **continuous budget** is a moving 12-month budget, adding a month as each month expires.



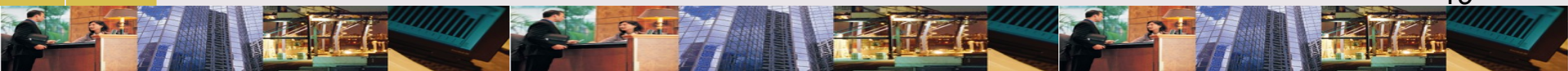
MASTER BUDGETS: Major Components

- ✱ Operating budget

 - ✱ Describes income generating activities of a firm

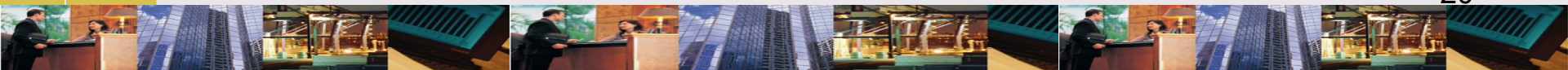
- ✱ Financial budgets

 - ✱ Detail inflows & outflows of cash



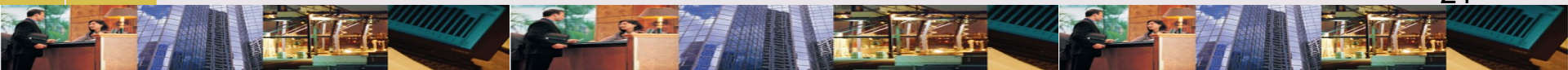
OPERATING BUDGETS: Steps in the Process

1. Sales budget
2. Production budget
3. Direct materials purchases budget
4. Direct labor budget
5. Overhead budget
6. Selling & administrative budget
7. Ending finished goods inventory budget
8. Cost of goods sold budget



TEXAS REX, INC.: Background

Texas Rex, Inc., is a trendy restaurant in the Southwest that sells T-shirts with a Texas Rex dinosaur logo. The operating budgets that follow are for manufacturing costs of Texas Rex T-shirts.



SALES BUDGET

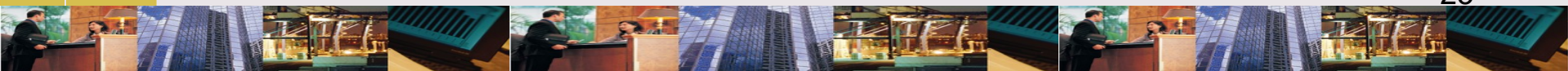
Schedule 1 describes
expected sales in
units & dollars.

Schedule 1 Texas Rex, Inc. Sales Budget For the Year Ended December 31, 2008					
	Quarter				Year
	1	2	3	4	
Units	1,000	1,200	1,500	2,000	5,700
Unit selling price	× \$10	× \$10	× \$10	× \$10	× \$10
Budgeted sales	<u>\$10,000</u>	<u>\$12,000</u>	<u>\$15,000</u>	<u>\$20,000</u>	<u>\$57,000</u>

FORMULAS: Production Units

Except for JIT systems, production budgets must meet sales needs & satisfy ending inventory requirements.

Units to be produced =
Expected unit sales +
Units in ending inventory –
Units in beginning inventory



PRODUCTION BUDGET

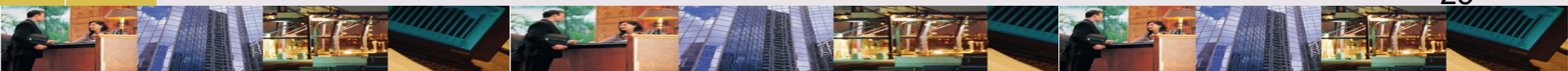
Schedule 2 describes
units to be produced
to meet Sales Budget.

Schedule 2 Texas Rex, Inc. Production Budget For the Year Ended December 31, 2008					
	Quarter				
	1	2	3	4	Year
Sales (Schedule 1)	1,000	1,200	1,500	2,000	5,700
Desired ending inventory	<u>240</u>	<u>300</u>	<u>400</u>	<u>200*</u>	<u>200</u>
Total needs	1,240	1,500	1,900	2,200	5,900
Less: Beginning inventory	<u>(180)</u>	<u>(240)</u>	<u>(300)</u>	<u>(400)</u>	<u>(180)</u>
Units to be produced	<u><u>1,060</u></u>	<u><u>1,260</u></u>	<u><u>1,600</u></u>	<u><u>1,800</u></u>	<u><u>5,720</u></u>

* Assume that sales for the first quarter of 2009 are estimated at 1,000 units.

TEXAS REX, INC.: Direct Materials

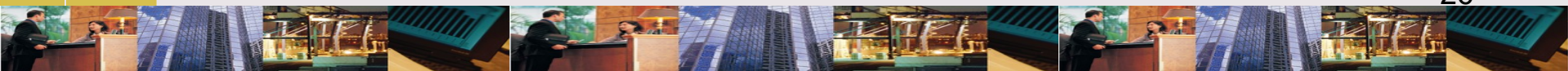
Texas Rex, Inc., purchases 2 direct materials (DM) for production of its Texas Rex T-shirts: plain T-shirts & ink to produce the dinosaur logo.



FORMULAS: Purchases

Direct materials purchases budget tells amount & cost of raw materials purchased in each period.

Direct materials (DM) purchased =
DM needed for production +
DM desired in ending inventory –
DM in beginning inventory



DM PURCHASES BUDGET

Schedule 3
Texas Rex, Inc.
Direct Materials Purchases Budget
For the Year Ended December 31, 2008

	Quarter				Year
Plain T-shirts:	1	2	3	4	
Units to be produced (Schedule 2)	1,060	1,260	1,600	1,800	5,720
Direct materials per unit	× 1	× 1	× 1	× 1	× 1
Production needs	1,060	1,260	1,600	1,800	5,720
Desired ending inventory	126	160	180	106*	106
Total needs	1,186	1,420	1,780	1,906	5,826
Less: Beginning inventory	(58)	(126)	(160)	(180)	(58)
Direct materials to be purchased	1,128	1,294	1,620	1,726	5,768
Cost per plain T-shirt	× \$3	× \$3	× \$3	× \$3	× \$3
Total purchase cost plain T-shirts	<u>\$3,384</u>	<u>\$3,882</u>	<u>\$4,860</u>	<u>\$5,178</u>	<u>\$17,304</u>

	Quarter				Year
Ink:	1	2	3	4	
Units to be produced (Schedule 2)	1,060	1,260	1,600	1,800	5,720
Direct materials per unit	× 5	× 5	× 5	× 5	× 5
Production needs	5,300	6,300	8,000	9,000	28,600
Desired ending inventory	630	800	900	530*	530
Total needs	5,930	7,100	8,900	9,530	29,130
Less: Beginning inventory	(390)	(630)	(800)	(900)	(390)
Direct materials to be purchased	5,540	6,470	8,100	8,630	28,740
Cost per ounce	× \$0.20	× \$0.20	× \$0.20	× \$0.20	× \$0.20
Total purchase cost of ink	<u>\$1,108</u>	<u>\$1,294</u>	<u>\$1,620</u>	<u>\$1,726</u>	<u>\$5,748</u>
Total direct materials purchase cost	<u>\$4,492</u>	<u>\$5,176</u>	<u>\$6,480</u>	<u>\$6,904</u>	<u>\$23,052</u>

*We do not know production for the first quarter of 2007 because we do not know sales for the second quarter of 2007. Therefore, the desired ending inventories of 106 plain T-shirts and 530 ounces of ink are given simply to complete this example.

Schedule 3 describes **DM** to be purchased to meet Production Budget.

DIRECT LABOR BUDGET

Schedule 4 shows hours & cost of **DL** needed to meet Production Budget.

Schedule 4
Texas Rex, Inc.
Direct Labor Budget
For the Year Ended December 31, 2008

	Quarter				
	1	2	3	4	Year
Units to be produced (Schedule 2)	1,060	1,260	1,600	1,800	5,720
Direct labor time per unit (hr.)	<u>× 0.12</u>	<u>× 0.12</u>	<u>× 0.12</u>	<u>× 0.12</u>	<u>× 0.12</u>
Total hours needed	127.2	151.2	192	216	686.4
Average wage per hour	<u>× \$10</u>	<u>× \$10</u>	<u>× \$10</u>	<u>× \$10</u>	<u>× \$10</u>
Total direct labor cost	<u>\$1,272</u>	<u>\$1,512</u>	<u>\$1,920</u>	<u>\$2,160</u>	<u>\$6,864</u>

OVERHEAD BUDGET

Schedule 5 shows expected **indirect costs** needed to meet Production Budget.

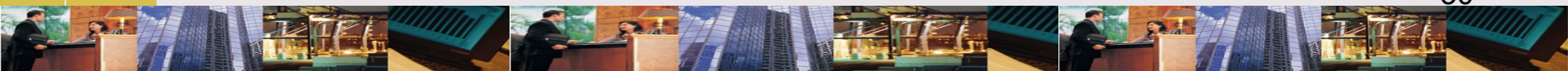
Schedule 5
Texas Rex, Inc.
Overhead Budget
For the Year Ended December 31, 2008

	Quarter				Year
	1	2	3	4	
Budgeted direct labor hours (Schedule 4)	127.2	151.2	192.0	216.0	686.4
Variable overhead rate	× \$5	× \$5	× \$5	× \$5	× \$5
Budgeted variable overhead	\$ 636	\$ 756	\$ 960	\$1,080	\$ 3,432
Budgeted fixed overhead*	<u>1,645</u>	<u>1,645</u>	<u>1,645</u>	<u>1,645</u>	<u>6,580</u>
Total overhead	<u>\$2,281</u>	<u>\$2,401</u>	<u>\$2,605</u>	<u>\$2,725</u>	<u>\$10,012</u>

*Includes \$540 of depreciation in each quarter.

How do we determine the
cost of finished goods
ending inventory?

Unit cost of finished goods
is (per unit) $DM + DL +$
Overhead.



FINISHED GOODS BUDGET

Schedule 6 shows unit **cost of finished goods** for balance sheet.

Schedule 6
Texas Rex, Inc.
Ending Finished Goods Inventory Budget
For the Year Ended December 31, 2008

Unit-cost computation:

Direct materials (\$3 + \$1)	\$4.00
Direct labor (0.12 hr. @ \$10)	1.20
Overhead:	
Variable (0.12 hr. @ \$5)	0.60
Fixed (0.12 hr. @ \$9.59)*	<u>1.15**</u>
Total unit cost	<u>\$6.95</u>

*Budgeted fixed overhead (Schedule 5)/Budgeted direct labor hours (Schedule 4) = \$6,580/686.4 = \$9.59**

**Rounded

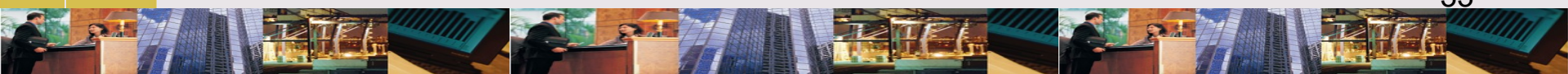
CGS BUDGET

Schedule 7 presents the expected **cost of goods sold** for the year.

Schedule 7 Texas Rex, Inc. Cost of Goods Sold Budget For the Year Ended December 31, 2008	
Direct materials used (Schedule 3) *	\$22,880
Direct labor used (Schedule 4)	6,864
Overhead (Schedule 5)	<u>10,012</u>
Budgeted manufacturing costs	\$39,756
Beginning finished goods	<u>1,251</u>
Goods available for sale	\$41,007
Less: Ending finished goods (Schedule 6)	<u>(1,390)</u>
Budgeted cost of goods sold	<u><u>\$39,617</u></u>
*Production needs = (5,720 plain T-shirts × \$3) + (28,600 oz. ink × \$0.20)	

How do we project income
from the operating
budgets?

Estimate selling &
administrative expenses, then
transfer all information into
projected income statement.



SALES & ADMINISTRATIVE EXPENSES BUDGET

Schedule 8 outlines planned expenditures for **nonmanufacturing activities**.

Schedule 8
Texas Rex, Inc.
Selling and Administrative Expenses Budget
For the Year Ended December 31, 2008

	Quarter				Year
	1	2	3	4	
Planned sales in units (Schedule 1)	1,000	1,200	1,500	2,000	5,700
Variable selling and administrative expenses per unit	× \$0.10	× \$0.10	× \$0.10	× 0.10	× \$0.10
Total variable expenses	<u>\$ 100</u>	<u>\$ 120</u>	<u>\$ 150</u>	<u>\$ 200</u>	<u>\$ 570</u>
Fixed selling and administrative expenses:					
Salaries	\$ 1,220	\$ 1,220	\$ 1,720	\$ 1,220	\$ 5,380
Utilities	50	50	50	50	200
Advertising	100	200	300	500	1,100
Depreciation	150	150	150	150	600
Insurance	200	200	200	200	800
Total fixed expenses	<u>\$ 1,720</u>	<u>\$ 1,820</u>	<u>\$ 2,420</u>	<u>\$ 2,120</u>	<u>\$ 8,080</u>
Total selling and administrative expenses	<u><u>\$ 1,820</u></u>	<u><u>\$ 1,940</u></u>	<u><u>\$ 2,570</u></u>	<u><u>\$ 2,320</u></u>	<u><u>\$ 8,650</u></u>

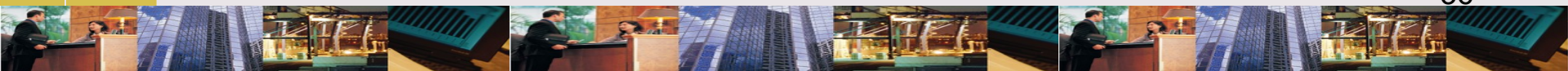
BUDGETED INCOME STATEMENT

Schedule 9 presents a projected income statement.

Schedule 9 Texas Rex, Inc. Budgeted Income Statement For the Year Ended December 31, 2008	
Sales (Schedule 1)	\$ 57,000
Less: Cost of goods sold (Schedule 7)	<u>(39,617)</u>
Gross margin	\$ 17,383
Less: Selling and administrative expenses (Schedule 8)	<u>(8,650)</u>
Operating income	\$ 8,733
Less: Interest expense (Schedule 10)	<u>(60)</u>
Income before income taxes	\$ 8,673
Less: Income taxes (Schedule 10)	<u>(2,550)</u>
Net income	<u><u>\$ 6,123</u></u>

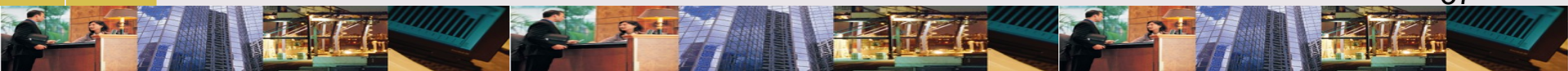
FINANCIAL BUDGETS

1. Cash budget
2. Budgeted balance sheet
3. Budget for capital expenditures



What is the purpose of the cash budget?

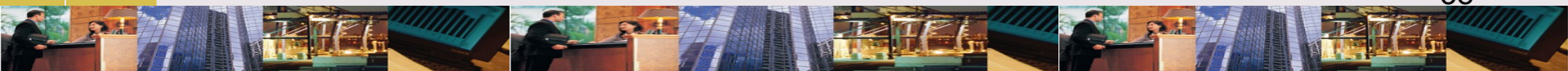
Cash budgets document the need for cash & the ability to repay debt.



FORMULA: Cash Budget

Projecting the ending cash balance includes cash collections, payments, & borrowings & includes minimum cash needed.

Ending cash balance =
Beginning balance +
(cash receipts – disbursements) +
(cash borrowing – repayments)



CASH BUDGET

Schedule 10
presents **projected**
cash needs.

Schedule 10 Texas Rex, Inc. Cash Budget For the Year Ended December 31, 2008						
	Quarter				Year	Source ^a
	1	2	3	4		
Beginning cash balance	\$ 5,200	\$ 1,023	\$ 1,611	\$ 3,762	\$ 5,200	g
Collections:						
Cash sales	2,500	3,000	3,750	5,000	14,250	b, 1
Credit sales:						
Current quarter	6,750	8,100	10,125	13,500	38,475	b, 1
Prior quarter	1,350	750	900	1,125	4,125	b, 1
Total cash available	<u>\$ 15,800</u>	<u>\$ 12,873</u>	<u>\$ 16,386</u>	<u>\$ 23,387</u>	<u>\$ 62,050</u>	
Less disbursements:						
Direct materials:						
Current quarter	\$ (3,594)	\$ (4,141)	\$ (5,184)	\$ (5,523)	\$(18,442)	c, 3
Prior quarter	(1,000)	(898)	(1,035)	(1,296)	(4,229)	c, 3
Direct labor	(1,272)	(1,512)	(1,920)	(2,160)	(6,864)	4
Overhead	(1,741)	(1,861)	(2,065)	(2,185)	(7,852)	d, 5
Selling and administrative	(1,670)	(1,790)	(2,420)	(2,170)	(8,050)	d, 8
Income taxes	—	—	—	(2,550)	(2,550)	f, 9
Equipment	(6,500)	—	—	—	(6,500)	e
Total disbursements	<u>\$(15,777)</u>	<u>\$(10,202)</u>	<u>\$(12,624)</u>	<u>\$(15,884)</u>	<u>\$(54,487)</u>	
Minimum cash balance	<u>(1,000)</u>	<u>(1,000)</u>	<u>(1,000)</u>	<u>(1,000)</u>	<u>(1,000)</u>	a
Total cash needs	<u><u>\$(16,777)</u></u>	<u><u>\$(11,202)</u></u>	<u><u>\$(13,624)</u></u>	<u><u>\$(16,884)</u></u>	<u><u>\$(55,487)</u></u>	
Excess (deficiency) of cash available over needs	\$ (977)	\$ 1,671	\$ 2,762	\$ 6,503	\$ 6,563	
Financing:						
Borrowings	1,000	—	—	—	1,000	a
Repayments	—	(1,000)	—	—	(1,000)	a
Interest ^b	—	(60)	—	—	(60)	a
Total financing	<u>\$ 1,000</u>	<u>\$ (1,060)</u>	<u>—</u>	<u>—</u>	<u>\$ (1,060)</u>	
Ending cash balance ^c	<u><u>\$ 1,023</u></u>	<u><u>\$ 1,611</u></u>	<u><u>\$ 3,762</u></u>	<u><u>\$ 6,503</u></u>	<u><u>\$ 6,503</u></u>	

^aLetters refer to the information on page 327. Numbers refer to schedules already developed.

^bInterest payment is $6/12 \times 0.12 \times \$1,000$. Since borrowings occur at the beginning of the quarter and repayments at the end of the quarter, the principal repayment takes place after six months.

^cTotal cash available minus total disbursements plus (or minus) total financing.

MASTER BUDGET INTERRELATIONSHIPS

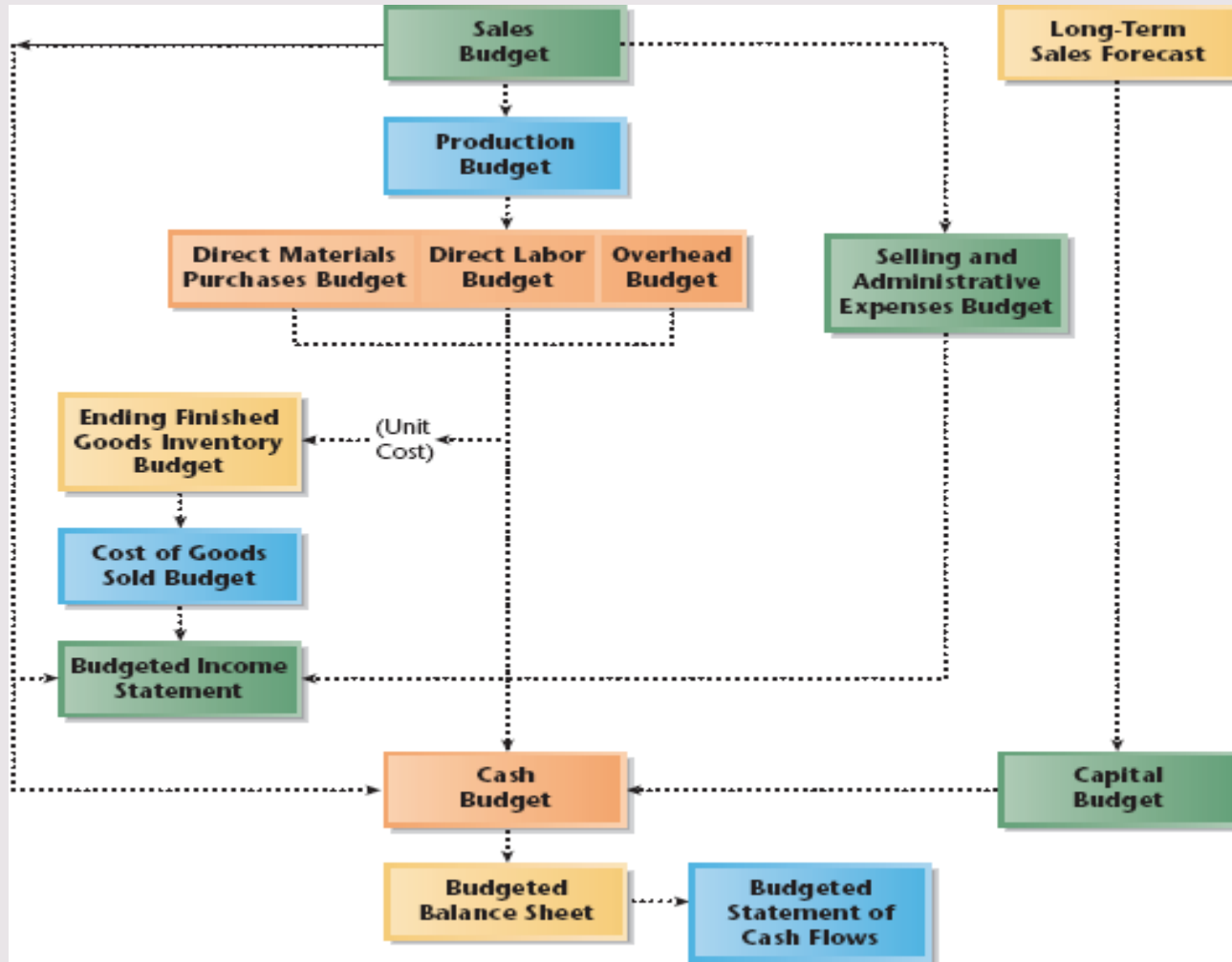


EXHIBIT 8-5

BUDGETED BALANCE SHEET

Schedule 11
Texas Rex, Inc.
Budgeted Balance Sheet
December 31, 2008

Assets

Current assets:		
Cash	\$ 7,503 ^a	
Accounts receivable	1,500 ^b	
Materials inventory	,424 ^c	
Finished goods inventory	<u>1,390^d</u>	
Total current assets		\$ 10,817
Property, plant, and equipment:		
Land	\$ 1,100 ^e	
Building and equipment	36,500 ^f	
Accumulated depreciation	<u>(7,760)^g</u>	
Total property, plant, and equipment		<u>29,840</u>
Total assets		<u><u>\$40,657</u></u>

Liabilities and Owners' Equity

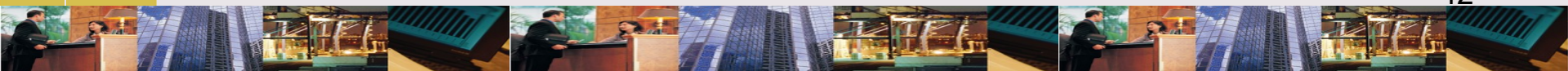
Current liabilities:		
Accounts payable		\$ 1,381 ^h
Owners' equity:		
Retained earnings	<u>\$39,276ⁱ</u>	
Total owners' equity		<u>39,276</u>
Total liabilities and owners' equity		<u><u>\$40,657</u></u>

Schedule 11
presents end of
year **balance sheet**.

LEARNING OBJECTIVE

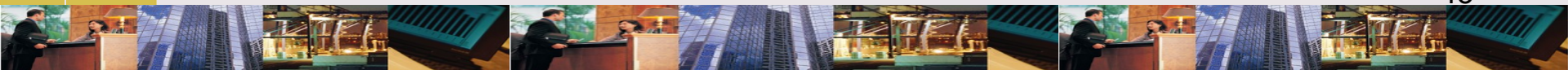
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Describe flexible budgeting & features that should encourage goal-congruent behavior.



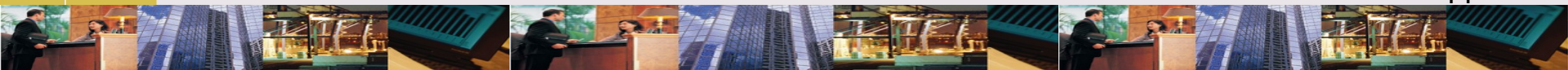
STATIC BUDGET: Definition

A budget for a particular level of activity.



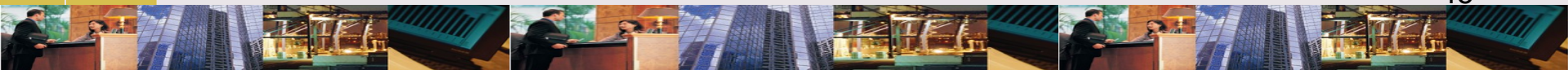
Why are static budgets not good for performance evaluation?

Actual level of activity may differ from the static budget level & misrepresent performance.



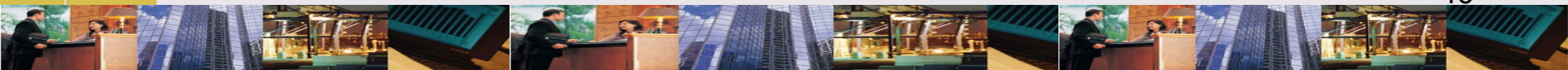
FLEXIBLE BUDGET: Definition

A budget for expected costs of a range of activity levels.



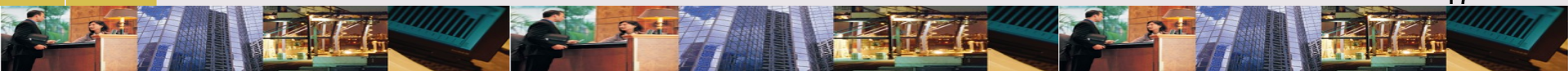
How are budgets related to performance evaluation?

Bonuses, salary increases, promotions are based on achieving or beating budget targets.



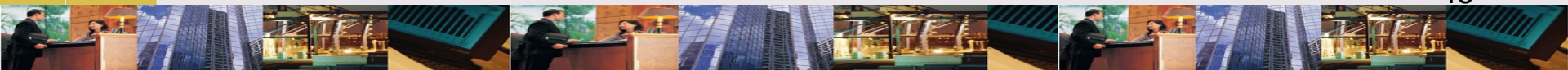
GOAL CONGRUENCE: Definition

Alignment of managerial & organizational goals.



What is participative budgeting?

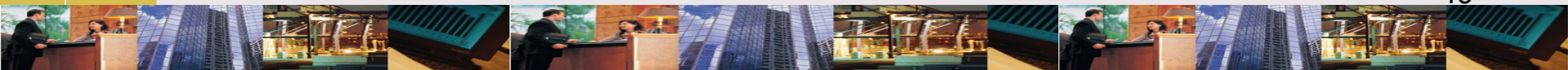
Participative budgeting involves subordinate managers in setting budget targets to achieve goal congruence.



PARTICIPATIVE BUDGETING

Potential problems

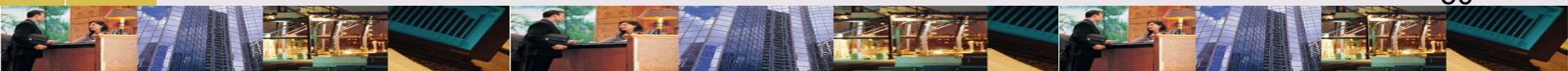
- ✧ Setting standards either too high or too low
- ✧ Building slack (padding) into the budget
 - ✧ Deliberately underestimating revenues, overestimating costs
- ✧ Pseudoparticipation



CONTROLLABLE COSTS:

Definition

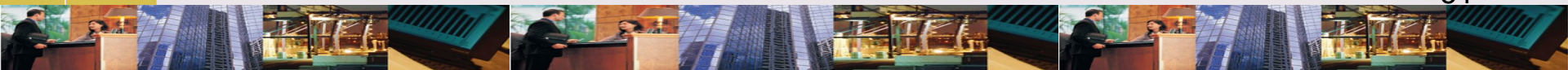
Are costs whose level a manager can influence



LEARNING OBJECTIVE

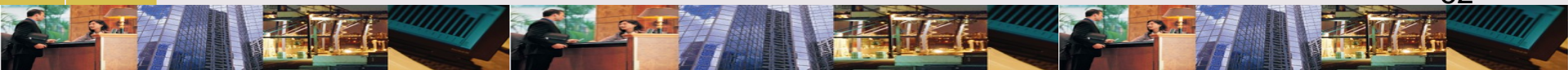
4

Explain how activity-based budgeting works.



ACTIVITY-BASED BUDGETING

Activity-based budgeting fits ABC & ABM systems. Budgets are developed for company activities to show the resources consumed. Can be done as a flexible budget.



ACTIVITY FLEXIBLE BUDGET

Budget can be developed based on different activity drivers.

Driver: Direct Labor Hours				
	Formula		Level of Activity	
	Fixed	Variable	10,000	20,000
Direct materials	—	\$10	\$100,000	\$200,000
Direct labor	—	8	80,000	160,000
Subtotal	<u>\$0</u>	<u>\$18</u>	<u>\$180,000</u>	<u>\$360,000</u>
Driver: Machine Hours				
	Formula		Level of Activity	
	Fixed	Variable	8,000	16,000
Maintenance	\$20,000	\$5.50	\$64,000	\$108,000
Machining	<u>15,000</u>	<u>2.00</u>	<u>31,000</u>	<u>47,000</u>
Subtotal	<u>\$35,000</u>	<u>\$7.50</u>	<u>\$95,000</u>	<u>\$155,000</u>
Driver: Number of Setups				
	Formula		Level of Activity	
	Fixed	Variable	25	30
Inspections	\$80,000	\$2,100	\$132,500	\$143,000
Setups	—	1,800	45,000	54,000
Subtotal	<u>\$80,000</u>	<u>\$3,900</u>	<u>\$177,500</u>	<u>\$197,000</u>
Driver: Number of Orders				
	Formula		Level of Activity	
	Fixed	Variable	15,000	25,000
Purchasing	<u>\$211,000</u>	<u>\$1</u>	<u>\$226,000</u>	<u>\$236,000</u>
Total			<u>\$678,500</u>	<u>\$948,000</u>

Exhibit 8-10 Activity Flexible Budget

EXHIBIT 8-10

CHAPTER 8

THE END